Lubomír Bureš

Nuclear engineer focused on computational modelling and technical problem solving.

me@lubomirbures.com || www.lubomirbures.com

www.linkedin.com/in/lubomir-bures || www.orcid.org/0000-0001-6204-0938

Professional Experience

Saltfoss Energy (formerly Seaborg Technologies) *Copenhagen, Denmark*
Senior Multiphysics Specialist March 2024 – present

- Collaborate as a technical lead on the Saltfoss molten-salt reactor design, guiding key architectural decisions and ensuring effective system integration.
- Address complex technical challenges in reactor physics, thermal hydraulics, and design optimisation.
- Develop, verify, and validate computational models for advanced nuclear applications.
- Publish research and present at scientific conferences to bolster the company's technical visibility.

Multiphysics Specialist October 2021 – March 2024

- Developed multiple computational models for molten-salt reactor analysis.
- Supervised a master thesis project with the candidate then joining the company as a full-time employee.
- Published peer-reviewed research and represented the company both in Denmark and internationally.

Paul Scherrer Institute *Villigen, Switzerland*
Doctoral Researcher October 2018 – September 2021

- Conducted independent doctoral research in computational physics.
- Designed and programmed software for high-fidelity numerical simulations applied to scientific research.
- Supervised four student projects, resulting in published research and successful student outcomes.
- Led exercise sessions for a course at ETH Zurich, improving student understanding.

Supplementary Projects

Winterlead *Remote*
Independent Consultant April 2024 – present

- Provide software development, computational engineering, and consultancy services on an ad hoc basis.
- Developed software solutions for Quantified Carbon.
- Volunteered technical expertise to the Repower Initiative.

Education

EPFL, Institute of Physics *Paul Scherrer Institute (PSI), Villigen, Switzerland*
Doctor of Science (PhD) 2018 – 2021

- *Degree awarded by*: École Polytechnique Fédérale de Lausanne (EPFL)
- *Thesis*: Fundamental study on microlayer dynamics in nucleate boiling
- *Achievements*: Passed without reserve (highest grade). Awarded the 2022 PhD Prize by the European Nuclear Education Network Association. Awarded the 2019 Fellowship for Research in Japan (cancelled due to Covid-19) by the Japanese Society for the Promotion of Science.

ETH Zurich – EPF Lausanne *Zürich & Lausanne, Switzerland*
Joint MSc in Nuclear Engineering 2016 – 2018

- *Thesis*: Direct numerical simulation of phase change in the presence of non-condensable gases
- *Achievements*: Graduated with distinction, overall GPA: 3.97/4.00 (third best among 1043 Master graduates). Awarded the EPFL Excellence Fellowship over the duration of the study programme.
- *Certifications*: Received the European Master of Science in Nuclear Engineering certification.
- *Internships*: National Cooperative for the Disposal of Radioactive Waste (Wettingen, Switzerland).
- *Extracurricular*: Represented the class in the EPFL student organisation (Fall semester 2016).

Czech Technical University *Prague, Czech Republic*
BSc in Nuclear Engineering 2013 – 2016

- *Thesis*: Application of calculation codes based on Monte-Carlo method for benchmark experiments
- *Achievements*: Graduated with honours, overall GPA: 3.94/4.00. Won a student thesis competition. Received the 2015 Talented Student Award from the Czech Technical University.
- *Internships*: Nuclear Physics Institute of the Czech Academy of Sciences (Řež, Czech Republic), Joint Institute for Nuclear Research (Dubna, Russia).

Skills

Languages	Czech native, English C2, German C1, Danish B2: Prøve i Dansk 3 – 8.5/12.
Programming	Python, C++, Modelica, MATLAB
Software Tools	Git, Jupyter, VS Code, Vim, MS Office, LaTeX
Operating Systems	Windows, Linux

Organisational Involvement

IDA Nuclear *Danish Society of Engineers, Denmark*
Board Member September 2025 – present

- Organise events promoting knowledge and education about nuclear and atomic technology.

Kildevæld Kulturcenter, Digitalt Værksted *Copenhagen, Denmark*
Volunteer / Superuser October 2025 – present

- Support open workshop sessions by helping citizens use digital fabrication tools, including 3D printing and laser cutting, and by sparring on project ideas.

PhD and Postdoc Association *Paul Scherrer Institute, Switzerland*
President October 2019 – May 2021

- Oversaw the creation of the association's bylaws, enabling its registration as a non-profit organisation.
- Led the association through the challenging times of the Covid-19 pandemic, ensuring continued support and engagement for members.
- Represented the association externally and within the institute, enhancing its visibility and influence.
- March 2019 to September 2019: Social Committee Head.

Various *Europe-wide*
Volunteer / Event Organiser October 2018 – November 2023

- Volunteered for and helped organise multiple climate-change-related events across Europe.
- Represented the Nuclear for Climate initiative at the UN Climate Change Conference COP25 in Madrid.